Task 3

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Part A

1. Show that eigenvectors corresponding to distinct eigenvalues of A are orthogonal. (4 marks)
2. Define... Show that $Q^TQ = I$. Briefly explain what this tells you about Q. (4 marks)
3. Using $Q^TQ = I$ and the following properties of determinants, show that det(Q) is either +1 or -1. (3 marks)
4. Consider the two orthogonal matrices. A data point x is transformed to z = Qx. (9 marks)
    1. Calculate the determinant of each matrix.
    2. Apply each transformation to the points... Sketch the original and transformed points.
    3. Describe each transformation geometrically. Relate your answers to the determinants.